

Fractions (Bhinn)

Hinglish Notes, Types, Operations aur Step-by-Step Solutions

Fraction kisi poori cheez ke hisse (part) ko represent karta hai. Is guide me fraction kya hai, uske types, simplest form, comparison aur sabhi operations (jod-ghata-guna-bhaag) simple Hinglish me samjhaye gaye hain, aur end me 22 practice questions ke STEP-BY-STEP solutions diye gaye hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

1. Fraction kya hota hai?

Definition Fraction kisi cheez ke barabar hisson me se kuch hisson ko batata hai. Ise p/q form me likha jata hai.

Example: $\frac{3}{4}$ ka matlab - poori cheez ko 4 barabar hisson me baanta, aur usme se 3 hisse liye.

Parts:

- Numerator (ansh) = upar wala number (kitne hisse liye). Jaise $\frac{3}{4}$ me 3.
- Denominator (har) = neeche wala number (total kitne hisse). Jaise $\frac{3}{4}$ me 4.
- Denominator kabhi 0 nahi ho sakta.

2. Fractions ke Types

EXAM * Exam me CONSTANTLY aata hai

- Proper Fraction: numerator < denominator. Jaise $\frac{3}{4}$, $\frac{2}{5}$.
- Improper Fraction: numerator $\geq$ denominator. Jaise $\frac{7}{4}$, $\frac{5}{5}$.
- Mixed Fraction: whole number + proper fraction. Jaise $1\frac{3}{4}$.
- Like Fractions: same denominator. Jaise $\frac{2}{5}$ aur $\frac{3}{5}$.
- Unlike Fractions: alag denominator. Jaise $\frac{1}{2}$ aur $\frac{1}{3}$.
- Equivalent Fractions: value barabar. Jaise $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$.
- Unit Fraction: numerator 1 ho. Jaise $\frac{1}{5}$, $\frac{1}{8}$.

3. Equivalent Fractions aur Simplest Form

EXAM * Exam me CONSTANTLY aata hai

Equivalent banane Numerator aur denominator dono ko same number se multiply ya divide karo. Jaise $\frac{1}{2} = \frac{(1 \times 2)}{(2 \times 2)} = \frac{2}{4}$.

Simplest Form Numerator aur denominator ko unke HCF se divide karo jab tak aur simplify na ho sake.

- Example: $\frac{6}{9} \rightarrow$ HCF 3 $\rightarrow \frac{(6/3)}{(9/3)} = \frac{2}{3}$ (simplest form).

4. Fractions ki Comparison

EXAM * Exam me CONSTANTLY aata hai

- Same denominator ho to bada numerator wala fraction bada hota hai. Jaise $\frac{3}{5} > \frac{2}{5}$.
- Alag denominator ho to LCM lekar same denominator banao, fir compare karo.
- Same numerator ho to chhota denominator wala bada hota hai. Jaise $\frac{1}{2} > \frac{1}{3}$.

5. Fractions par Operations

EXAM * Exam me CONSTANTLY aata hai

(a) Addition / Subtraction Same denominator ho to numerators jodo/ghatao, denominator wahi rakho. Alag ho to pehle LCM se same banao.

- Example: $1/2 + 1/3 \rightarrow \text{LCM } 6 \rightarrow 3/6 + 2/6 = 5/6$

(b) Multiplication Numerator x numerator, denominator x denominator. Fir simplify karo.

- Example: $2/3 \times 3/4 = 6/12 = 1/2$

(c) Division Doosre fraction ko ulta (reciprocal) karke multiply karo.

- Example: $1/2 \div 1/4 = 1/2 \times 4/1 = 4/2 = 2$

6. Conversions

Improper $\rightarrow$ Mixed Numerator ko denominator se divide karo. Quotient = whole part, remainder/denominator = fraction part.

- Example: $7/2 \rightarrow 7 \div 2 = 3 \text{ remainder } 1 \rightarrow 3 \frac{1}{2}$

Mixed $\rightarrow$ Improper (whole x denominator) + numerator, denominator wahi.

- Example: $2 \frac{1}{3} \rightarrow (2 \times 3) + 1 = 7 \rightarrow 7/3$

Fraction $\rightarrow$ Decimal Numerator ko denominator se divide karo. Jaise $1/4 = 0.25$.

6B. Reciprocal aur Ascending/Descending Order

EXAM * Exam me CONSTANTLY aata hai

Reciprocal (ulta) Fraction ko ulta kar do - numerator aur denominator swap. Jaise $3/4$ ka reciprocal $4/3$.

- Fraction aur uske reciprocal ka product hamesha 1 hota hai: $3/4 \times 4/3 = 1$.
- 1 ka reciprocal 1 hai; $5 (=5/1)$ ka reciprocal $1/5$ hai.

Ascending / Descending Pehle saare fractions ko same denominator (LCM) me badlo, fir numerators dekh kar order lagao.

- Example: $1/2, 1/3, 1/4$ ko ascending me $\rightarrow \text{LCM } 12 \rightarrow 6/12, 4/12, 3/12 \rightarrow 1/4 < 1/3 < 1/2$.

Quick Revision

Proper: upar < neeche ($3/4$) | Improper: upar $\geq$ neeche ($7/4$)

Add/Subtract: denominator same karo (LCM).

Multiply: seedha upar-upar, neeche-neeche.

Divide: doosre ko ulta karke multiply.

Simplest form: HCF se divide.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain, taaki har type ka method clear ho jaye (jod, ghata, guna, bhaag, negative, mixed).

Example 1: $2/3 + 1/4$ (unlike addition)

- Step: Denominator alag hain $\rightarrow \text{LCM}(3, 4) = 12$.
- Step: Dono ko 12 wale denominator me badlo:
- Step: $2/3 = 8/12$ (upar-neeche $\times 4$), $1/4 = 3/12$ ($\times 3$).
- Step: Numerators jodo: $8/12 + 3/12 = 11/12$.

Answer: $11/12$

Example 2: $\frac{5}{6} - \frac{1}{4}$ (unlike subtraction)

- Step: LCM(6, 4) = 12.
- Step: $\frac{5}{6} = \frac{10}{12}$ (x2), $\frac{1}{4} = \frac{3}{12}$ (x3).
- Step: Ghatao: $\frac{10}{12} - \frac{3}{12} = \frac{7}{12}$.

Answer: $\frac{7}{12}$

Example 3: $\frac{2}{3} \times \frac{4}{5}$ (multiplication)

- Step: Upar x upar: $2 \times 4 = 8$.
- Step: Neeche x neeche: $3 \times 5 = 15$.
- Step: = $\frac{8}{15}$. HCF(8,15)=1, isliye already simplest.

Answer: $\frac{8}{15}$

Example 4: $\frac{2}{4} \times \frac{6}{5} \times -\frac{2}{7}$ (negative, 3 fractions)

- Step: Sign dekho: ek hi negative hai -> answer NEGATIVE hoga.
- Step: Numerators multiply: $2 \times 6 \times 2 = 24$.
- Step: Denominators multiply: $4 \times 5 \times 7 = 140$.
- Step: = $-\frac{24}{140}$.
- Step: Simplify: HCF(24,140) = 4 -> $-\frac{24}{4} / \frac{140}{4}$.

Answer: $-\frac{6}{35}$

Example 5: $\frac{3}{4} \div \frac{2}{5}$ (division)

- Step: Division me doosre ka reciprocal lo: $\frac{2}{5} \rightarrow \frac{5}{2}$.
- Step: Ab multiply: $\frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$.
- Step: $\frac{15}{8}$ improper hai -> mixed = $1 \frac{7}{8}$.

Answer: $\frac{15}{8}$ (ya $1 \frac{7}{8}$)

Example 6: $1 \frac{1}{2} + 2 \frac{1}{3}$ (mixed addition)

- Step: Pehle improper banao: $1 \frac{1}{2} = \frac{3}{2}$, $2 \frac{1}{3} = \frac{7}{3}$.
- Step: LCM(2,3) = 6 -> $\frac{3}{2} = \frac{9}{6}$, $\frac{7}{3} = \frac{14}{6}$.
- Step: Jodo: $\frac{9}{6} + \frac{14}{6} = \frac{23}{6}$.
- Step: Mixed me: $\frac{23}{6} = 3 \frac{5}{6}$.

Answer: $\frac{23}{6}$ (ya $3 \frac{5}{6}$)

Example 7: $\frac{18}{24}$ ko simplest form me likho

- Step: HCF(18, 24) = 6 nikalo.
- Step: Upar-neeche ko 6 se divide: $\frac{18}{6} = 3$, $\frac{24}{6} = 4$.

Answer: $\frac{3}{4}$

Example 8: $\frac{3}{4}$ aur $\frac{5}{6}$ me bada kaunsa? (compare)

- Step: LCM(4, 6) = 12.
- Step: $\frac{3}{4} = \frac{9}{12}$, $\frac{5}{6} = \frac{10}{12}$.
- Step: $\frac{10}{12} > \frac{9}{12}$.

Answer: $\frac{5}{6}$ bada hai

Example 9: $-\frac{3}{5} \times 10$ (negative x whole number)

- Step: 10 ko fraction maano: $\frac{10}{1}$.
- Step: $-\frac{3}{5} \times \frac{10}{1} = -\frac{30}{5}$.
- Step: $-\frac{30}{5} = -6$.

Answer: -6

Example 10: 20 ka $\frac{3}{4}$ (kisi quantity ka fraction)

- Step: 'ka' / 'of' ka matlab multiply hota hai.

- Step: $\frac{3}{4} \times 20 = \frac{3}{4} \times \frac{20}{1} = \frac{60}{4}$.
- Step: $\frac{60}{4} = 15$.

Answer:15

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1.** $\frac{3}{4}$ me numerator kaunsa number hai?
- Q2.** $\frac{7}{4}$ kis type ka fraction hai (proper / improper)?
- Q3.** Kya $\frac{2}{3}$ aur $\frac{4}{6}$ equivalent fractions hain?
- Q4.** $\frac{5}{10}$ ko simplest form me likho.
- Q5.** $\frac{1}{2} + \frac{1}{2} = ?$
- Q6.** $\frac{3}{5} + \frac{1}{5} = ?$
- Q7.** $\frac{1}{2} + \frac{1}{3} = ?$
- Q8.** $\frac{3}{4} - \frac{1}{4} = ?$
- Q9.** $\frac{2}{3} \times \frac{3}{4} = ?$
- Q10.** $\frac{1}{2} \div \frac{1}{4} = ?$
- Q11.** $\frac{7}{2}$ ko mixed fraction me likho.
- Q12.** $2\frac{1}{3}$ ko improper fraction me likho.
- Q13.** $\frac{1}{4}$ ko decimal me likho.
- Q14.** Proper fraction ki definition kya hai?
- Q15.** Unit fraction kya hota hai? Ek example do.
- Q16.** Kya $\frac{4}{8}$, $\frac{1}{2}$ aur $\frac{3}{6}$ sab equivalent hain?
- Q17.** $\frac{2}{5}$ aur $\frac{3}{5}$ me se bada kaunsa hai?
- Q18.** $\frac{1}{2}$ aur $\frac{1}{3}$ me se bada kaunsa hai?
- Q19.** $\frac{6}{9}$ ka simplest form kya hai?
- Q20.** $\frac{3}{4} \times 4 = ?$
- Q21.** Mixed fraction kya hota hai? Ek example do.
- Q22.** 0.5 ko fraction me likho.

Step-by-Step Solutions

- Q1.** $\frac{3}{4}$ me numerator?
- Step: Fraction me upar wala number numerator hota hai.
 - Step: $\frac{3}{4}$ me upar 3 hai.
- Answer:**Numerator = 3
- Q2.** $\frac{7}{4}$ ka type?
- Step: Numerator = 7, Denominator = 4.
 - Step: Yahan numerator (7) > denominator (4).
- Answer:**Improper Fraction
- Q3.** $\frac{2}{3}$ aur $\frac{4}{6}$ equivalent?
- Step: $\frac{2}{3}$ ke upar-neeche ko 2 se multiply karo.
 - Step: $(2 \times 2) / (3 \times 2) = \frac{4}{6}$.

- Step: Dono ki value barabar nikli.

Answer:Haan, equivalent hain

Q4. 5/10 simplest form?

- Step: HCF(5, 10) = 5 nikalo.
- Step: Upar-neeche ko 5 se divide: $(5/5)/(10/5)$.

Answer:1/2

Q5. $1/2 + 1/2 = ?$

- Step: Denominator same (2) hai, to numerators jodo.
- Step: $(1+1)/2 = 2/2$.
- Step: $2/2$ simplify $\rightarrow 1$.

Answer:1

Q6. $3/5 + 1/5 = ?$

- Step: Denominator same (5) hai.
- Step: Numerators jodo: $(3+1)/5$.

Answer:4/5

Q7. $1/2 + 1/3 = ?$

- Step: Denominator alag hain $\rightarrow$ LCM(2,3) = 6.
- Step: $1/2 = 3/6$ aur $1/3 = 2/6$.
- Step: Ab jodo: $3/6 + 2/6$.

Answer:5/6

Q8. $3/4 - 1/4 = ?$

- Step: Denominator same (4) hai.
- Step: Numerators ghatao: $(3-1)/4 = 2/4$.
- Step: Simplify: HCF 2 se $\rightarrow 1/2$.

Answer: $2/4 = 1/2$

Q9. $2/3 \times 3/4 = ?$

- Step: Multiplication: upar x upar, neeche x neeche.
- Step: $(2 \times 3)/(3 \times 4) = 6/12$.
- Step: Simplify: HCF 6 se $\rightarrow 1/2$.

Answer: $6/12 = 1/2$

Q10. $1/2 \div 1/4 = ?$

- Step: Division me doosre ka reciprocal lo: $1/4 \rightarrow 4/1$.
- Step: Ab multiply: $1/2 \times 4/1 = 4/2$.
- Step: $4/2$ simplify $\rightarrow 2$.

Answer:2

Q11. $7/2 \rightarrow$ mixed fraction?

- Step: Numerator ko denominator se divide: $7 \div 2$.
- Step: Quotient = 3, Remainder = 1.
- Step: Mixed = quotient + (remainder/denominator).

Answer:3 $1/2$

Q12. $2 \frac{1}{3} \rightarrow$ improper fraction?

- Step: Formula: (whole x denominator) + numerator.
- Step: $(2 \times 3) + 1 = 7$.
- Step: Denominator wahi (3) rakho.

Answer:7/3

Q13. $1/4 \rightarrow$ decimal?

- Step: Numerator ko denominator se divide: $1 \div 4$.

Answer: 0.25

Q14. Proper fraction definition?

- Step: Jis fraction me numerator chhota ho denominator se.

Answer: Numerator < Denominator (jaise $3/4$)

Q15. Unit fraction kya hai?

- Step: Wo fraction jiska numerator 1 ho.

Answer: Jaise $1/5$, $1/7$

Q16. $4/8$, $1/2$, $3/6$ equivalent?

- Step: $4/8$ ko HCF 4 se simplify $\rightarrow 1/2$.
- Step: $3/6$ ko HCF 3 se simplify $\rightarrow 1/2$.
- Step: Teeno ki value $1/2$ nikli.

Answer: Haan, sab equivalent

Q17. $2/5$ aur $3/5$ me bada?

- Step: Denominator same (5) hai.
- Step: Bada numerator wala bada: $3 > 2$.

Answer: $3/5$ bada hai

Q18. $1/2$ aur $1/3$ me bada?

- Step: Numerator same (1) hai.
- Step: Chhota denominator wala bada: $2 < 3$.

Answer: $1/2$ bada hai

Q19. $6/9$ simplest form?

- Step: $HCF(6, 9) = 3$ nikalo.
- Step: Upar-neeche ko 3 se divide: $(6/3)/(9/3)$.

Answer: $2/3$

Q20. $3/4 \times 4 = ?$

- Step: 4 ko fraction maano: $4/1$.
- Step: $3/4 \times 4/1 = 12/4$.
- Step: Simplify: $12/4 = 3$.

Answer: 3

Q21. Mixed fraction kya hai?

- Step: Whole number + proper fraction milkar banta hai.

Answer: Jaise $2 \frac{1}{3}$

Q22. $0.5 \rightarrow$ fraction?

- Step: $0.5 = 5/10$ likho.
- Step: HCF 5 se simplify: $(5/5)/(10/5)$.

Answer: $1/2$